



KULIYAH OF ENGINEERING

DEEP LEARNING (MCTA 4363)

AI DUO : AI MATHEMATICAL OLYMPIAD

TRACK 1

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1.0 ITERATIVE MODEL DEVELOPMENT LOG

1.1 DATASET CONSTRUCTION AND PREPARATION

Objective : Build a unified mathematical reasoning dataset using multiple Hendrycks MATH subjects.

Steps :

1. Loaded seven Hendrycks subject datasets:
 - A. Algebra
 - B. Counting and Probability
 - C. Geometry
 - D. Intermediate Algebra
 - E. Number Theory
 - F. Prealgebra
 - G. Precalculus
2. Concatenated all subjects into one combined dataset (7500 training samples).
3. Added an original_index column for traceability.
4. Saved processed dataset to Google Drive.

Outcome : A clean, multi-domain math dataset ready for SFT training.

1.2 TRAIN AND TEST SPLIT

Training Samples : 6750

Test Samples : 750

Method : `train_test_split(test_size=0.1)`

1.3 BASE MODEL SELECTION AND LOADING

Model Used : Qwen/Qwen2.5-Math-1.5B

Quantization :

- A. LoRA (4-bit NF4 quantization)
- B. `load_in_4bit=True`
- C. `bnb_4bit_use_double_quant=True`
- D. `bnb_4bit_compute_dtype=torch.float16`

Reason : This allows efficient training on a single Colab T4 GPU.

1.4 TOKENIZATION CONFIGURATION

- A. max_length=512
- B. truncation=True
- C. padding="max_length"
- D. Labels mirror input IDs (standard SFT objective)

1.5 LoRA FINE TUNING

Applied low-rank adaptation to reduce memory use and training cost.

LoRA Settings :

- A. r=8
- B. lora_alpha=32
- C. lora_dropout=0.05
- D. target_modules=["q_proj", "v_proj"]

Training Arguments :

- A. batch_size=1
- B. gradient_accumulation=4
- C. num_train_epochs=2
- D. learning_rate=2e-5
- E. fp16=True
- F. Evaluation at each epoch
- G. Model saved after each epoch

1.6 TRAINING RESULTS

Epoch	Train Loss	Validation Loss
1	0.243	0.269
2	0.267	0.266

Interpretation :

1. Loss stabilized after two epochs.

2. No signs of divergence.
3. Model successfully learned mathematical syntax and reasoning structure.

1.8 EVALUATION

Method :

1. Generate solution using a structured prompt.
2. Extract final answer using regex (boxed{...}).
3. Compare predicted vs ground-truth answer.

This evaluation measures exact-answer correctness, not reasoning quality.

1.9 TECHNIQUES

Experiment	Model	Dataset	Epochs	LR	Batch Size	Accuracy
Baseline	Qwen2.5-Math-1.5B	Hendrycks	2	2e-5	1	60%
Experiment 2	Prompt Optimization	Hendrycks	2	2e-5	1	X%
Experiment 3	IMO Fine-Tuning	Hendrycks + IMO	1	1e-5	1	X%

2.0 COMPARATIVE BENCHMARK PERFORMANCE

To evaluate the effectiveness of the proposed mathematical reasoning system, a controlled benchmark assessment was conducted using a fixed set of 185 algebra-focused questions derived from the Hendrycks MATH dataset. This benchmark was specifically designed to ensure consistency and fairness by testing both the baseline model and future fine-tuned versions on the exact same unseen problem set.

Before Fine-Tuning

The baseline evaluation was performed using the pretrained Qwen2.5-Math-1.5B model without any domain-specific fine-tuning. Under these conditions, the model achieved:

FINAL PERFORMANCE SUMMARY	
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Dataset Used	: Hendrycks MATH (Algebra Only)
Total Questions Tested	: 185
Correct Answers	: 14
Incorrect Answers	: 171
Final Accuracy	: 7.57%

These results establish the initial benchmark performance of the raw pretrained model. While the baseline model demonstrated some foundational mathematical reasoning capability, its relatively low accuracy indicates limited specialization for Olympiad-level algebraic problem solving. The findings highlight the necessity for domain adaptation techniques such as LoRA fine-tuning, prompt optimization, and dataset specialization to significantly enhance reasoning precision and problem-solving effectiveness.

After Fine-Tuning

The next phase of development will involve applying supervised fine-tuning using the Hendrycks MATH training dataset combined with LoRA-based parameter-efficient adaptation. This stage is expected to improve:

- Step-by-step reasoning consistency
- Mathematical derivation quality
- Final answer formatting reliability
- Overall benchmark accuracy

Following successful fine-tuning, the model will be re-evaluated using the same fixed algebra benchmark to allow direct before-and-after performance comparison. This iterative methodology ensures that all performance gains can be scientifically attributed to the fine-tuning process rather than differences in dataset selection or evaluation conditions.

Comparative Development Objective

The primary objective of this benchmarking framework is not only to measure raw accuracy improvements, but also to systematically validate whether domain-specific fine-tuning enhances the model's ability to solve competition-level mathematical problems more effectively than the pretrained baseline.

This comparative evaluation approach strengthens the project's scientific rigor by providing:

- Controlled experimentation
- Reproducible benchmarking
- Quantifiable performance improvements
- Clear iterative development evidence

Overall, this benchmark serves as the foundational performance reference point from which all future optimization strategies including prompt engineering, advanced dataset specialization, and hyperparameter refinement will be measured.

3.0 CONCLUSION

In conclusion, this project has established the initial foundation for developing an AI Mathematical Olympiad system through dataset preparation, baseline benchmarking, and model implementation. While the current results provide an important starting point, there are still several improvements required to enhance overall performance, including fine-tuning optimization, prompt engineering, and specialized dataset integration. Future development will focus on refining the system's mathematical reasoning accuracy, solution quality, and competitive benchmark performance through continuous iterative experimentation and model enhancement.

4.0 GITHUB LINK

<https://github.com/khalisahamani/AI-Duo>